SAT Math

Equivalent Expressions 2

Question #	ID		Answer
2.1	dd4ab4c4	$4a^2+20ab+25b^2$ Which of the following is a factor of the polynomial above? A. $a+b$ B. $2a+5b$ C. $4a+5b$ D. $4a+25b$	B
2.2	b8caaf84	If $p=3x+4$ and $v=x+5$, which of the following is equivalent to $pv-2p+v$? A. $3x^2+12x+7$ B. $3x^2+14x+17$ C. $3x^2+19x+20$ D. $3x^2+26x+33$	B
2.3	ad2ec615	Which of the following is equivalent to the expression x^4-x^2-6? A. $(x^2+1)(x^2-6)$ B. $(x^2+2)(x^2-3)$ C. $(x^2+3)(x^2-2)$ D. $(x^2+6)(x^2-1)$	B

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2.4	42c71eb5	$(2x+5)^2-(x-2)+2(x+3)$	A

Which of the following is equivalent to the expression above?

A. $4x^2+21x+33$

B. $4x^2+21x+29$

C. $4x^2+x+29$

D. $4x^2+x+33$

2.5	a05bd3a4	C

Which of the following expressions is equivalent to x^2-5 ?

A. $(x+\sqrt{5})^2$

B. $(x-\sqrt{5})^2$

C. $(x+\sqrt{5})(x-\sqrt{5})$

D. $(x+5)(x-1)$

2.6	cc776a04	C

Which of the following is an equivalent form of $(1.5x-2.4)^2-(5.2x^2-6.4)$?

A. $-2.2x^2+1.6$

B. $-2.2x^2+11.2$

C. $-2.95x^2-7.2x+12.16$

D. $-2.95x^2-7.2x+0.64$

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Question #	ID		Answer
2.7	a520ba07	$\sqrt[3]{x^3y^6}$ Which of the following expressions is equivalent to the expression above? A. y^2 B. xy^2 C. y^3 D. xy^3	B
2.8	5b6af6b1	Which expression is equivalent to $(d-6)(8d^2-3)$? A. $8d^3-14d^2-3d+18$ B. $8d^3-17d^2+48$ C. $8d^3-48d^2-3d+18$ D. $8d^3-51d^2+48$	C
2.9	a255ae72	If $x^2=a+b$ and $y^2=a+c$, which of the following is equal to $(x^2-y^2)^2$? A. $a^2-2ac+c^2$ B. $b^2-2bc+c^2$ C. $4a^2-4abc+c^2$ D. $4a^2-2abc+b^2c^2$	B

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Question #	ID		Answer
2.10	463eec13	If $x \neq 0$, which of the following expressions is equivalent to $\frac{\sqrt{16x^4y^8}}{x^3}$? A. $8x^2y^4$ B. $4xy^4$ C. $4x^{-2}y^2$ D. $4x^{-1}y^4$	D
2.11	a1bf1c4e	x^2+6x+4 Which of the following is equivalent to the expression above? A. $(x+3)^2+5$ B. $(x+3)^2-5$ C. $(x-3)^2+5$ D. $(x-3)^2-5$	B
2.12	f237ccfc	The sum of $-2x^2+x+31$ and $3x^2+7x-8$ can be written in the form ax^2+bx+c, where a, b, and c are constants. What is the value of $a+b+c$?	32
2.13	a391ed22	$\left(\frac{1}{2}x+\frac{3}{2}\right)\left(\frac{3}{2}x+\frac{1}{2}\right)$ The expression above is equivalent to ax^2+bx+c, where a, b, and c are constants. What is the value of b?	2.5

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Question #	ID		Answer
2.14	c3a72da5	Which of the following is equivalent to the sum of $3x^4+2x^3$ and $4x^4+7x^3$? A. $16x^{14}$ B. $7x^8+9x^6$ C. $12x^4+14x^3$ D. $7x^4+9x^3$	D
2.15	16de54c7	$2x^2+5x-12$ If the given expression is rewritten in the form $(2x-3)(x+k)$, where k is a constant, what is the value of k ?	4
2.16	d9137a84	Which expression represents the product of $\left(x^{-6}y^3z^5\right)$ and $\left(x^4z^5+y^8z^{-7}\right)$? A. $x^{-2}z^{10}+y^{11}z^{-2}$ B. $x^{-2}z^{10}+x^{-6}z^{-2}$ C. $x^{-2}y^3z^{10}+y^8z^{-7}$ D. $x^{-2}y^3z^{10}+x^{-6}y^{11}z^{-2}$	D
2.17	3e9cc0c2	Which of the following is equivalent to $(1-p)(1+p+p^2+p^3+p^4+p^5+p^6)$? A. $1-p^8$ B. $1-p^7$ C. $1-p^6$ D. $1-p^5$	B

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Question #	ID		Answer
2.18	7348f046	$(2x+3)-(x-7)$ Which of the following is equivalent to the given expression? A. $x-4$ B. $3x-4$ C. $x+10$ D. $2x^2+21$	C
2.19	b47419f4	$\left(\frac{1}{2}x+3\right)-\left(\frac{2}{3}x-5\right)$ Which of the following is equivalent to the expression above? A. $-\frac{1}{6}x+8$ B. $-\frac{1}{6}x-2$ C. $-\frac{1}{3}x^2+\frac{1}{2}x+15$ D. $-\frac{1}{3}x^2-\frac{9}{2}x-15$	A
2.20	8838a672	$(4x^3-5x^2+3)-(6x^3+2x^2-x)$ Which of the following expressions is equivalent to the expression above? A. $-10x^3-3x^2+x+3$ B. $-2x^3-7x^2+x+3$ C. $-2x^3-3x^2+x+3$ D. $10x^3-7x^2-x+3$	B

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Question #	ID		Answer
2.21	0b3d25c5		D

Which of the following is equivalent to $\sqrt[4]{x^2+8x+16}$, where $x>0$?

A. $(x+4)^4$

B. $(x+4)^2$

C. $(x+4)$

D. $(x+4)^{\frac{1}{2}}$

2.22 c602140f — Answer: B

$$(x-11y)(2x-3y)-12y(-2x+3y)$$

Which of the following is equivalent to the expression above?

A. $x-23y$

B. $2x^2-xy-3y^2$

C. $2x^2+24xy+36y^2$

D. $2x^2-49xy+69y^2$

2.23 3206b905 — Answer: D

Which of the following expressions is equivalent to $8x^{10}-8x^9+88x$?

A. $x(7x^{10}-7x^9+87x)$

B. $x(8^{10}-8^9+88)$

C. $8x(x^{10}-x^9+11x)$

D. $8x(x^9-x^8+11)$

2.24 26eb61c1 — Answer: C

Which expression is equivalent to $6x^8y^2+12x^2y^2$?

A. $6x^2y^2(2x^6)$

B. $6x^2y^2(x^4)$

C. $6x^2y^2(x^6+2)$

D. $6x^2y^2(x^4+2)$

Question #	ID		Answer
2.25	6d04c89d	The expression $\frac{24}{6x+42}$ is equivalent to $\frac{4}{x+b}$, where *b* is a constant and $x > 0$. What is the value of *b*?	A

A. 7

B. 10

C. 24

D. 252